Differential Expression Part 3

- 1-page summary of your proposed project due on March 3rd
- Today's lab activity is 07_DE_DESeq2_limma.qmd

DESeq2 Analysis Pipeline Overview

1) Load count data (column 1 = gene names, subsequent columns are counts for samples, row names are headings)

2) Create “DESeqDataSet” for your samples that assigns metadata to counts (contains the contrasts of interest, similar to the “contrast matrix” that is used by edgeR)
    a) main difference is there is a common reference by default (intercept)

3) Filter out genes with low counts

4) Perform all normalization steps and differential expression testing (combines all of median-of-ratios normalization for library size and composition with estimating dispersions)

5) Write table of outputs (logFC, p-values, FDR-adjusted p-values)

limma voom Analysis Pipeline Overview

Very similar to edgeR (same group created both)

1) Load count data (column 1 = gene names, subsequent columns are counts for samples, row names are headings)

2) Create “groups” for your samples that you want to compare (e.g., group 1 = WT unstressed, group 2 = WT EtOH stressed, group 3 = mutant unstressed, group 4 = mutant stressed)

3) Create a “design matrix” that assigns groups to the samples in your count table

4) Create a “contrast matrix” for your comparisons of interest (e.g., group 1 (WT unstressed) vs group 2 (WT EtOH stressed))

limma voom Analysis Pipeline Overview

5) Filter out genes with low counts

6) Normalize to total reads per sample and composition (TMM)

7) Convert counts to logCPM

8) Apply voom transformation to logCPM data (estimates the mean-variance relationship of the log-counts, generates a precision weight for each observation)

9) Fit a linear model to each gene (lmFit function)

10) Use empirical Bayes to “squeeze” variance to towards a common value and perform a moderated F test

11) Write table of outputs (logFC, p-values, FDR-adjusted p-values)

Comparison Across Methods

Method	Normalization	Model	Test
edgeR	TMM	Negative binomial GLM	Quasi-likelihood F test
DESeq2	Median-of-ratios	Negative binomial GLM	Wald test (or LRT)
limma-voom	TMM + voom weights	Linear model on log-CPM	Moderated F test

- edgeR/DESeq2: Model counts directly with negative binomial distribution
- limma-voom: Transform counts to log scale, model with normal distribution using precision weights